Up-type FCNC in presence of Dark Matter

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ABSTRACT: Dark Matter (DM) is a known unknown. Apart, current experimental constraints on flavor-changing neutral current (FCNC) processes involving up-type quarks also provide scope to explore physics beyond the Standard Model (SM). In this article, we establish a connection between the flavor sector and the DM sector with minimal extension of the SM. Here a singlet complex scalar field, stable under $\mathcal{Z}_3$ symmetry, acts as DM and couples to SM up-type quarks through a heavy Dirac vector-like quark (VLQ), which shares the same $\mathcal{Z}_3$ charge as of the DM. The model thus addresses the observed $D^0-\bar{D}^0$ mixing, top-FCNC interactions, and D^0 meson decays, together with DM relic density, while evading the direct and indirect DM search bounds. The model can be probed at the future high-energy muon collider, through distinctive signatures of VLQ production, where the VLQ decays into DM and SM particles, abiding by the existing bounds.

Contents

1 Introduction

The Standard Model (SM) of particle physics has been proved to be an appropriate theory to describe fundamental particles and interactions to a great extent. Amongst its shortcomings, processes involving flavor-changing neutral current (FCNC) indicate towards physics beyond the Standard Model (BSM) [1–3]. Presence of non-luminous dark matter (DM) [4], which contributes significantly ($\sim 26.8\%$) [5] to the energy budget of the universe, is another challenge, as SM doesn't contain any such particle. Despite astrophysical and cosmological evidences [6–11], DM is still undiscovered. Some characteristics to associate DM are its non-zero mass, lack of electromagnetic interactions, and dynamical stability over cosmological timescales, consistent with the predictions of the Lambda Cold Dark Matter (ΛCDM) paradigm [12–14]. Among the leading CDM candidates, classified by their production mechanisms and interaction strengths, are Weakly Interacting Massive Particles (WIMPs) [15], Feebly Interacting Massive Particles (FIMPs) [16], and Strongly Interacting Massive Particles (SIMPs) [17], amongst others. DM searches are going on through direct detection (DD) [18–20], indirect detection (ID) [21–23], and collider experiments [24–26].

Null results have consequently resulted in exclusion bounds on pertinent observables, which serve as the dominant constraints in exploring DM phenomenology.

FCNC processes are powerful indirect probes of BSM physics due to their strong suppression in the SM. While flavor physics has primarily focused on the down-type quark sector, the up sector remains less explored, yet offers a theoretically clean environment to test BSM effects. Charm meson processes, such as those involving the D^0 meson, have their short-distance contributions suppressed by loop-level dynamics and CKM factors, making them largely dominated by long-distance effects [27]. If $D^0-\bar{D}^0$ mixing originates entirely from BSM contributions, it imposes stringent constraints on New Physics (NP) scenarios [28]. Leptonic decays of D^0, where both long- and short-distance effects are suppressed, have been studied model-independently [29, 30] and can be significantly enhanced in NP models with extended Higgs sectors or leptoquarks [31–33]. Similarly, top-quark FCNC decays like $t \to u_j X$ ($X = \gamma, g, Z, H$) are extremely rare in the SM, offering a clean channel to search for BSM signals. Recent experimental advances have notably improved our sensitivity to such rare processes, making the up-type quark sector an increasingly important frontier for NP searches [34, 35].

In this work, a potential correlation between DM and FCNC observables is investigated. Our model involves a complex scalar dark matter (CSDM) stabilized by $\mathcal{Z}_3$ symmetry [36, 37], which helps in semi annihilation [38], as opposed to a real scalar stabilized by a $\mathcal{Z}_2$ symmetry [39]. Such processes provide additional depletion channels to satisfy relic density, evading DD bounds, and ID bounds. After strong constraints on minimal scalar and fermion DM models, such extensions are phenomenologically motivated. Apart, we have a Dirac vector-like quark (VLQ) that interacts exclusively with the right-handed up-type quarks[1], due to its specific $\mathrm{U}(1)_{\mathtt{Y}}$ hypercharge assignment, $\mathtt{Y} = 2/3$. This provides further annihilation and co-annihilation channels of DM.

Due to strong interaction, the vector-like quark (VLQ) is expected to yield a significant cross-section at the LHC. The absence of a signal excess thus constrains the VLQ to a high-mass regime. The VLQ decays to up-type quark and DM, providing $t\bar{c} + \not{E}$ signal as a potential probe for the model. Multi-TeV muon colliders offer a promising platform to explore such heavy VLQs, while extensive studies have been done at both hadron [43–50] and lepton colliders [51, 52].

This paper is organized as follows: Section 2 introduces the VLQ model that links the dark sector to the Standard Model. In Section 3, we discuss various experimental constraints on the model. Section 4 presents our analysis of FCNC effects, dark matter phenomenology, and search strategies for future colliders. Finally, Section 5 provides a summary and conclusion.

[1]For a vector-like lepton extension of the complex scalar DM model, see [40]. For scenarios involving a Majorana fermion ψ, refer to [35, 41, 42]. In such cases, the $\mathcal{Z}_3$ symmetry cannot be employed for DM stabilization; instead, a $\mathcal{Z}_2$ symmetry must be utilized.

2 Model framework

In this article, we present a minimal extension of a complex scalar singlet DM model by incorporating a Dirac VLQ, which carries $\mathrm{U(1)_Y}$ hypercharge $\mathtt{Y} = 2/3$[2] and couples only to the right-handed up-type quarks of the SM. The SM extended Lagrangian is given by[3],

$$\begin{aligned}\mathcal{L} \supset & -\mu_H^2|H|^2 - \lambda_{\mathrm{H}}|H|^4 + |\partial_\mu\Phi|^2 - m_\Phi^2|\Phi|^2 - \lambda_\Phi|\Phi|^4 - \frac{\mu_3}{2}\left[\Phi^3 + (\Phi^*)^3\right] \\ & - \lambda_{\Phi\mathrm{H}}|\Phi|^2\left(H^\dagger H - v^2/2\right) + \overline{\psi}[i\gamma^\mu\left(\partial_\mu + ig_s T^a G^a_\mu + ig'\mathtt{Y}B_\mu\right)]\psi \\ & - m_\psi\overline{\psi}\psi - \left(\mathrm{y_u}\overline{\psi}u_R\Phi + \mathrm{y_c}\overline{\psi}c_R\Phi + \mathrm{y_t}\overline{\psi}t_R\Phi + \mathrm{h.c.}\right) .\end{aligned} \tag{2.1}$$

Here, the parameters associated with the kinetic term of ψ carry their usual meanings, and $\mu_H^2 < 0$. The model is perturbative if $\lambda_\Phi \leq \pi$, $\lambda_{\mathrm{H}} \leq 4\pi$ [54] and $\{\mathrm{y}_c,\ \mathrm{y}_t,\ \mathrm{y}_u\} < \sqrt{4\pi}$, while the unitarity bounds are always weaker than this. To keep the potential bounded from below, we choose the quartic interaction in such a way that it satisfies the vacuum stability conditions, $\lambda_H > 0,\ \lambda_\Phi > 0,\ \lambda_{\Phi\mathrm{H}} + 2\sqrt{\lambda_\Phi\lambda_{\mathrm{H}}} > 0$. Moreover, the maximum allowed value of the cubic parameter, μ_3 is approximately max $\mu_3 \approx 2\sqrt{2}\sqrt{\frac{\lambda_\Phi}{\delta}}\, m_\Phi$, while $\delta = 2$ gives the absolute stability bound [36]. The SM and BSM fields relevant for our analysis are their respective quantum numbers are detailed in Tab. 1.

Fields	$\mathrm{SU(3)_c}$	$\mathrm{SU(2)_L}$	$\mathrm{U(1)_Y}$	$\mathcal{Z}_3$
H	1	2	1/2	1
u_R, c_R, t_R	3	1	2/3	1
Φ	1	1	0	ω/ω^2
ψ	3	1	2/3	ω/ω^2

Table 1: The model particle contents and their corresponding quantum numbers.

The free parameters of this model are

$$\{m_\Phi,\ m_\psi,\ \mathrm{y_u},\ \mathrm{y_c},\ \mathrm{y_t},\ \lambda_{\Phi\mathrm{H}},\ \mu_3\}\,. \tag{2.2}$$

These serve as the basis for analyzing the relic density of DM and its detection prospects, as well as for exploring various flavor physics phenomena involving up-type quarks.

3 Experimental constraints

3.1 Flavor Changing Processes

In this work, we have considered a fermionic mediator χ interacting with the complex scalar DM Φ as well as the SM fermions via the Yukawa portal, as can be seen from Eq. (2.1).

[2] Alternatively, the scalar can acquire hypercharge, enabling the singlet fermion to serve as DM [53].

[3] The source of origin of low energy quark portal renormalizable interaction term, $\overline{\psi}u_R\Phi$, could be a dimension-5 effective operator $(C_u/\Lambda)\,\overline{\Psi}H\Phi u_R$, where $\Psi = (\psi^0\ \psi)^{\mathrm{T}}$ is a vector like Quark doublet, and also transform similarly under $\mathcal{Z}_3$ like Φ. After EWSB, obtain a term like $\left(C_u v/\sqrt{2}\Lambda\right)\overline{\psi}u_R\Phi \equiv \mathrm{y}_u\overline{\psi}u_R\Phi$.

The hypercharge of the VLQ ψ allows it to only interact with the up-type quarks u, c, t. Constraints from the low-energy process related to the charm and up-quark and high-energy processes related to the top quark can be derived, which can put bounds on the allowed couplings and masses of the NP fields. The important processes of low-energy are meson mixing: $D^0 - \bar{D}^0$, rare decay: $D^0 \to \ell^+\ell^-$, semi-leptonic decays: $D^0 \to (\pi^0, \rho^0, \omega)\ell\bar{\ell}$ etc. The important high-energy processes are the rare decays of top quark like: $t \to c(u)X$ with $X \in \gamma,\ g,\ Z$ and h. Following, we will discuss our model's impact on these processes.

3.1.1 Meson mixing

The $D^0 - \bar{D}^0$ mixing is an important probe to study the NP, since it is highly suppressed in the SM via CKM and loop contributions. The meson mixing observable Δm is defined as

$$\Delta m = \frac{|\mathcal{M}|}{m_D}\,. \tag{3.1}$$

Here, $\mathcal{M}$ is the amplitude of the meson mixing process. The experimental value is measured by LHCb [55], BaBar [56], Belle [57], CDF [58]. In our analysis, we have used the value averaged by HFLAV [59], which is given as:

$$\Delta m = (0.997 \pm 0.116) \times 10^{10}\,\mathrm{s}^{-1}. \tag{3.2}$$

This mixing is primarily influenced by long-distance amplitudes, which are challenging to compute. The SM short-distance effect is very suppressed of the order of $\sim \mathcal{O}(10^6)\,\mathrm{s}^{-1}$ [60]. Assuming that the observed mixing arises solely from non-SM contributions, one can derive strong constraints on potential NP models [28, 61]. In our work, we have also considered that the effect is entirely dominated by the NP effect. In our case, we get the mixing via one-loop box diagrams as shown in the Fig. 1.

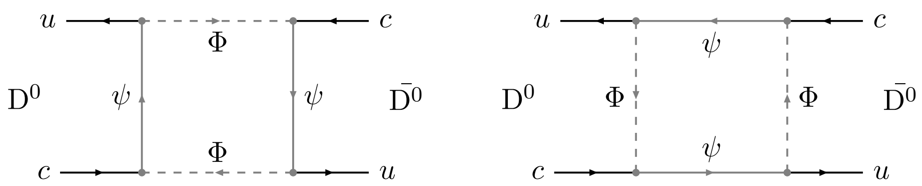


Figure 1: The relevant Feynman diagrams contributing to the process $D^0 - \bar{D}^0$ mixing.

The loop contributions can be written as:

$$\mathcal{L}_{\text{mixing}} = \mathcal{C}_{RR}[\bar{u}\gamma_\mu P_R c]\,[\bar{u}\gamma^\mu P_R c]\,, \tag{3.3}$$

with $\mathcal{C}_{RR}$ being the loop contribution. Here, we only get a non-negligible contribution to the right-handed vector current. As mentioned, since the contribution from the SM short-distance effect is highly suppressed, we assume the entire effect is coming via NP. The expression for the mass difference Δm_{NP} is given by:

$$\Delta m_{\text{NP}} = \frac{2}{3}\, B_D\, m_D\, f_D^2\, \mathcal{C}_{RR}\,, \tag{3.4}$$

$B_2 = 0.757$ [62] is the bag factor coming from:

$$\langle D^0 | (\bar{u}\gamma_\mu P_R c)\,(\bar{u}\gamma^\mu P_R c) | \bar{D}^0 \rangle = \frac{2}{3}\, B_D\, f_D^2 m_D^2\,, \tag{3.5}$$

with $f_D = 212.0$ MeV, being the decay constant of the D^0 meson [63]. Another way to express the mixing observable is through the ratio of the NP contribution to the SM contribution, as shown in [64, 65]. This is typically done to eliminate uncertainties from decay constants or bag parameters. In our case, since the SM contribution to the process is negligible, we simply take Δm as the observable. The allowed parameter space from $D^0 - \bar{D}^0$ mixing will be discussed in the next section.

3.1.2 Rare decays

Another significant avenue for probing NP is through rare decay processes, as they are highly suppressed in the SM due to loop suppression and the GIM mechanism. For charm mesons, this suppression is even stronger. Most charm processes are dominated by long-distance effects. Similar to the case of meson mixing, the SM contributions to rare decays are several orders of magnitude smaller than experimental results, leading to the assumption that the entire contribution comes from BSM physics. In SM, the branching ratio to muon channel is of the order of $\sim \mathcal{O}(10^{-18})$ [66] and the long-distance effect $\sim \mathcal{O}(10^{-13})$. The mass of D^0 meson allows it to decay to muon and electron pairs. The corresponding experimental branching ratios are given by:

$$\mathcal{B}(D^0 \to e^+\, e^-) < 7.9 \times 10^{-8} \quad [67], \tag{3.6}$$

$$\mathcal{B}(D^0 \to \mu^+\, \mu^-) < 3.1 \times 10^{-9} \quad [68]. \tag{3.7}$$

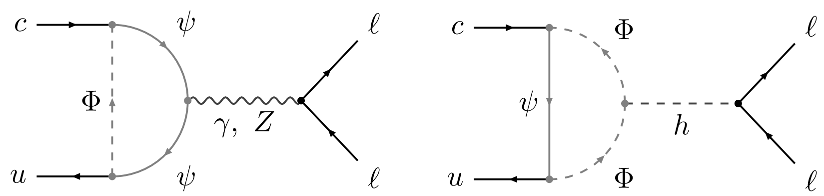


Figure 2: Feynman diagrams contributing to the rare decay of charm meson $D^0 \to \ell\bar{\ell}$.

The effective Hamiltonian describing the decay is given by:

$$\mathcal{H}_{\text{eff}}^{c\to u\ell\ell} = -\frac{4G_F}{\sqrt{2}}\lambda_q \sum_{i=1,\ldots,10,S^{(\prime)},P^{(\prime)},T,T5} \mathcal{C}_i \mathcal{O}_i\,, \tag{3.8}$$

λ_q being the CKM element $V_{cq}^* V_{uq}$. $q = (b, s, d)$,depending on the particle in the loop. The contributing operators have a structure [29]:

$$\mathcal{O}_S = \frac{g_e^2}{16\pi^2}\,(\bar{c}P_R u)\,(\bar{\ell}\;\ell), \qquad \mathcal{O}_S' = \frac{g_e^2}{16\pi^2}\,(\bar{c}P_L u)\,(\bar{\ell}\;\ell)\,, \tag{3.9}$$

$$\mathcal{O}_P = \frac{g_e^2}{16\pi^2}\,(\bar{c}P_R u)\,(\bar{\ell}\gamma_5\ell), \qquad \mathcal{O}_S' = \frac{g_e^2}{16\pi^2}\,(\bar{c}P_L u)\,(\bar{\ell}\gamma_5\ell)\,, \tag{3.10}$$

$$\mathcal{O}_{10} = -\frac{\alpha_e}{4\pi}(\bar{u}_L\gamma^\mu c_L)(\bar{\ell}\gamma_\mu\gamma_5\ell), \qquad \mathcal{O}_{10}' = -\frac{\alpha_e}{4\pi}(\bar{u}_R\gamma^\mu c_R)(\bar{\ell}\gamma_\mu\gamma_5\ell)\,. \tag{3.11}$$

The CKM elements are embedded within the Wilson coefficients, which for short-distance effects are detailed in [29]. Here, $D^0 - \bar{D}^0$ mixing arises from one-loop box diagrams, while the decay $D^0 \to \ell\bar{\ell}$ occurs via one-loop penguin diagrams, as shown in Fig. 2. The branching ratio can be written as:

$$\mathcal{B}(D^0 \to \ell\bar{\ell}) = \tau_D \frac{G_F^2 \alpha_e^2}{64\pi^3} f_D^2 m_D^5 \beta(m_D^2) \left(|P|^2 + \beta(m_D^2)|S|^2 \right) , \tag{3.12}$$

with,

$$S = \frac{1}{m_c} \left(C_S - C_S' \right) , \tag{3.13a}$$

$$\beta(q^2) = \sqrt{1 - 4m_\ell^2/q^2} , \tag{3.13b}$$

$$P = \frac{1}{m_c}(C_P - C_P') + \frac{2m_\ell}{m_D^2}(C_{10} - C_{10}') . \tag{3.13c}$$

In our work, we obtain contributions to the Wilson operators $C_{10}^{(\prime)}$ and $C_S^{(\prime)}$. Since the loop contributions to the $c \to u\ell\bar{\ell}$ process are extremely small, we do not derive any effective constraints on the couplings from this process, and thus it will be disregarded in the combined analysis. Similarly, we also do not obtain significant contributions from other semi-leptonic D-meson processes, such as $D^0 \to (\pi^0, \rho^0, \omega)\ell\bar{\ell}$, $D_s^+ \to K^+\ell\bar{\ell}$, $D^0 \to \pi^0\nu\bar{\nu}$, and other observables related to $c \to u\ell\bar{\ell}$ and $c \to u\nu\bar{\nu}$ transitions.

Furthermore, CP-violating decays of the D-meson are also expected to receive NP contributions in our model. One such observable is the difference in CP asymmetries in $D^0 \to K^+K^-$ and $D^0 \to \pi^+\pi^-$ decays. The experimental value of the observable ΔA_{CP} is given by LHCb [69] as:

$$\Delta A_{\rm CP}^{\rm exp} = (-15.4 \pm 2.9) \times 10^{-4} . \tag{3.14}$$

In contrast, the SM contribution is significantly smaller, estimated as $\Delta A_{\rm CP}^{\rm SM} = (2.0 \pm 0.3) \times 10^{-5}$. The decays $D^0 \to K^+K^-$ and $D^0 \to \pi^+\pi^-$ in our model are governed by gluon-mediated penguin processes. Furthermore, since all the couplings in our models are assumed to be real, the resulting contributions, even when accounting for hadronic enhancement, remain insufficient to satisfy the experimental value of $\Delta A_{\rm CP}$ within the viable parameter space. For more details on the contribution to $\Delta A_{\ rmCP}$ while considering a Dirac fermion in a t-channel DM model can be found in [70] while analysis involving a Majorana fermion is discussed in ref. [42]. Further studies involving a Z' mediator are discussed in [71]. However, the invisible decay of the D^0 meson, $\mathcal{B}(D^0 \to \text{invisible}) < 9.4 \times 10^{-5}$ [72, 73], could be relevant for low-mass DM phenomenology where ($\sum_i m_{{\rm DM}_i} < m_{D^0} \approx 1.864$ GeV) [74].

3.1.3 Top-FCNC decays

The branching ratios of top-FCNC processes are highly suppressed within the SM, while experimental measurements report values several orders of magnitude larger (see Table 2), indicating a potential manifestation of NP. In this model, we have Yukawa interactions

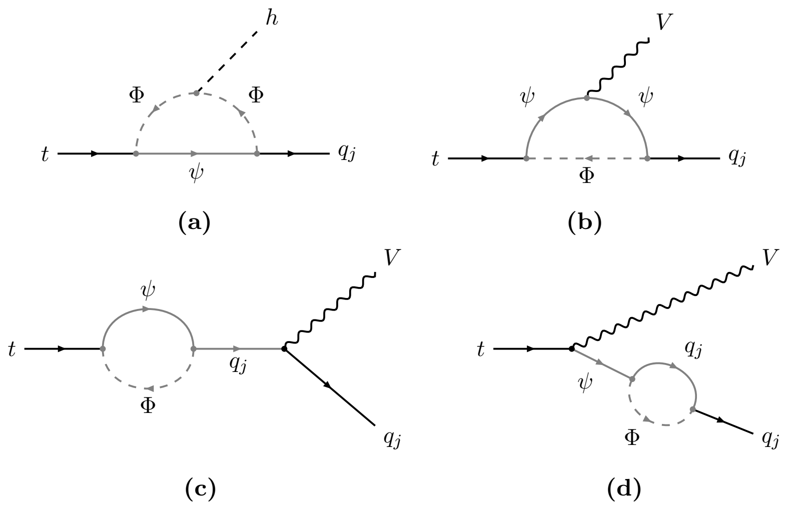


Figure 3: Feynman diagrams that are contributing to $t \to q_j\ V\ (h)$, where $q_j \in \{c,\ u\}$ and $V \in \{\gamma,\ Z,\ g\}$.

with the up-type quarks, as can be seen from Eq. (2.1), which can have an impact on the FCNC decays of top quark such as $t \to c(u)X$, where $X = \{\gamma, g, Z, h\}$. Therefore, we get top-FCNC processes via one-loop penguin diagrams as can be seen from Fig. 3. The

Branching Ratio	SM [75]	Experimental bound	Ref
$\mathrm{BR}(t \to ch)$	3×10^{-15}	3.7×10^{-4}	CMS [76]
$\mathrm{BR}(t \to uh)$	2×10^{-17}	1.9×10^{-4}	
$\mathrm{BR}(t \to cg)$	5×10^{-12}	3.7×10^{-4}	ATLAS [77]
$\mathrm{BR}(t \to ug)$	4×10^{-14}	6.1×10^{-5}	
$\mathrm{BR}(t \to c\gamma)$	5×10^{-14}	$4.2\ (4.5) \times 10^{-5}$ LH (RH)	ATLAS [78]
$\mathrm{BR}(t \to u\gamma)$	4×10^{-16}	$8.5\ (12) \times 10^{-6}$ LH (RH)	
$\mathrm{BR}(t \to cZ)$	1×10^{-14}	$1.3\ (1.2) \times 10^{-4}$ LH (RH)	ATLAS [79]
$\mathrm{BR}(t \to uZ)$	7×10^{-17}	$6.2\ (6.6) \times 10^{-5}$ LH (RH)	

Table 2: Available SM predictions and experimental bounds of the top-FCNC decays by considering left (right) handed effective couplings denoted by LH (RH).

general Lagrangian responsible for different top-FCNC decays can be given by:

$$\begin{aligned}\mathcal{L}_{t\to q_j X} = \ & -\bar{q}\gamma_\mu \left(C^{VL}_{tq_jZ} P_L + C^{VR}_{tq_jZ} P_R\right) t Z^\mu - \bar{q} i \sigma_{\mu\nu} \epsilon^{\mu *}_Z p^\nu \left(C^{TL}_{tq_jZ} P_L + C^{TR}_{tq_jZ} P_R\right) t \\ & -\bar{q} i \sigma_{\mu\nu} \epsilon^{\mu *}_\gamma p^\nu \left(C^{L}_{tq_j\gamma} P_L + C^{R}_{tq_j\gamma} P_R\right) t - \bar{q}\left(C^{L}_{tq_jh} P_L + C^{R}_{tq_jh} P_R\right) t h\,. \end{aligned} \tag{3.15}$$

with $q_j = c, u$. The expression for the branching ratios can be given by:

$$\mathcal{B}(t \to q_j Z) = \frac{\tau_t}{32\pi m_t} \frac{(m_t^2 - m_Z^2)^2}{m_t^2 m_Z^2} \Big\{ (m_t^2 + 2m_Z^2) \left((C^L_{tq_jZ})^2 + (C^R_{tq_jZ})^2 \right)$$

$$-6 m_t m_Z^2 \left(C^{VL}_{tq_jZ} C^{TR}_{tq_jZ} + C^{VR}_{tq_jZ} C^{TL}_{tq_jZ} \right) \tag{3.16}$$

$$+ m_Z^2 (m_t^2 + m_Z^2) \left((C^{TL}_{tq_jZ})^2 + (C^{TR}_{tq_jZ})^2 \right) \Big\} \,, \tag{3.17}$$

$$\mathcal{B}(t \to q_j h) = \frac{\tau_t m_t}{32\pi} \left(1 - \frac{m_h^2}{m_t^2} \right)^2 \left(|C^L_{tq_jh}|^2 + |C^R_{tq_jh}|^2 \right) . \tag{3.18}$$

The branching ratios to the photon and gluon are suppressed in our case; therefore, their explicit expressions are not presented. The corresponding experimental measurements and SM predictions for each decay are summarized in Table 2.

3.2 Dark Matter

In accordance with Eq. (2.1) and Table 1, the complex scalar field Φ is identified as the only viable DM candidate among the fields Φ and ψ. Since both fields transform under the same $\mathcal{Z}_3$ symmetry, the stability of Φ is kinematically guaranteed by the mass hierarchy $m_\Phi < m_\psi$. Nevertheless, the heavier dark sector particle ψ can still contribute to the DM relic abundance through co-annihilation processes, as depicted in Fig. 18. The mass of thermal DM is constrained by various theoretical and observational studies, providing the following limits [80–83]:

$$10 \text{ MeV} \lesssim m_{\rm DM} \lesssim 100 \text{ TeV} . \tag{3.19}$$

The presence of the Higgs portal interaction in the CSDM model, where the Higgs decays into invisible particles, imposes constraints on the coupling $\lambda_{\Phi \rm H}$. The 95% confidence level (CL) upper limit set on the branching fraction of the 125 GeV Higgs boson decay into invisible particles at $\sqrt{s} = 13$ TeV by the ATLAS and CMS detectors at the LHC is given by [84–86]:

$$\mathcal{B}(h \to \text{inv}) < \begin{cases} 0.107\,, & \text{ATLAS}\,, \\ 0.150\,, & \text{CMS}\,. \end{cases} \tag{3.20}$$

Alternatively, the invisible width of the Z boson, using an integrated luminosity of 37 (36.3) fb^{-1}, produced in proton-proton collisions at a center-of-mass energy of 13 TeV at the ATLAS (CMS) detector, is given by [87, 88]:

$$\Gamma(Z \to \text{inv}) < \begin{cases} 506 \text{ MeV}\,, & \text{ATLAS}\,, \\ 523 \text{ MeV}\,, & \text{CMS}\,. \end{cases} \tag{3.21}$$

The WMAP Collaboration constraint on the DM relic density, inferred from Planck – 2018 measurements, is given by [5]:

$$\Omega_{\rm DM} h^2 = 0.1200 \pm 0.0012 . \tag{3.22}$$

The most recent results from the search for nuclear recoil induced by WIMPs using the LUX-ZEPLIN (LZ) two-phase xenon time projection chamber set the strongest spin-independent (SI) exclusion limit on the WIMP-nucleon cross-section, $\sim 2.1 \times 10^{-48}$ cm^2 at the 90% CL for a DM mass of 36 GeV. Although several experiments, such as LUX-ZEPLIN [89], XENONnT [90], PandaX-4T [91], and PandaX-xT [92], also place limits on the DM-nucleon scattering cross-section, these limits are less stringent. In this framework, DM-nucleon scattering is governed by inelastic s-channel and t-channel processes, as shown in Fig. 4a and Fig. 4b, respectively. Interestingly, the DM-nucleon scattering cross-section depends on $\mathrm{y_u}$ and $\lambda_{\Phi\mathrm{H}}$, along with the m_ψ and m_Φ masses, which are also strongly constrained by the DM relic density and FCNC observables.

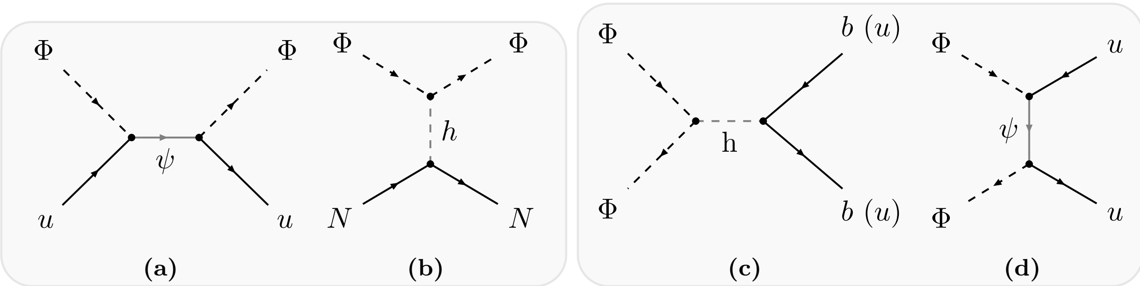


Figure 4: The Feynman diagrams are related to the direct (4a, 4b) and indirect detection (4c, 4d) of WIMP.

Gamma-rays might be produced at the Galactic Center through the annihilation of DM, as constrained by observations from Fermi-LAT [93] and projected results [94]. In our model, the presence of the Higgs portal coupling and the direct coupling with up-type quarks predominantly affects the DM annihilation into $b\bar{b}$ (Fig. 4c) and $u\bar{u}$ (Fig. 4d). Notably, these coupling parameters ($\mathrm{y_u}$ and $\lambda_{\Phi\mathrm{H}}$) are constrained by both the DM relic density (which requires a larger value to obtain the correct relic density) and DD constraints (which require a smaller value to remain consistent with current bounds and be within the reach of future DD experiments) when m_Φ and m_ψ are $\mathcal{O}(\mathrm{GeV} - \mathrm{TeV})$ range. Simultaneously, the magnitude of these couplings is further restricted by ID constraints from Fermi-LAT. The direct and indirect limits on the relic-allowed parameter space are discussed in Section 4.2.

3.3 VLQ search at the LHC

VLQs are predicted in many extensions of the Standard Model, but dedicated searches at the LHC remain limited. Most studies in this direction have focused on scenarios where VLQs mix with SM quarks, leading to their decay into SM quarks and gauge bosons [44, 95]. While such scenarios provide explicit signatures of VLQs, their final states bear little resemblance to the case at hand, where the VLQ decays into an SM quark and dark matter. A more comparable scenario arises in supersymmetric quark partner (*squark*) searches, where SUSY particles decay into SM quarks and neutralinos (invisible particles), closely mirroring our scenario. In this section, we recast an existing ATLAS *stop* (supersymmetric top partner) search analysis [96] from LHC Run II at 13 TeV with an integrated luminosity of 139 ab^{-1}. This analysis considers a *stop* pair decaying into a top quark and a neutralino, resulting in a final state with a top-quark pair and missing transverse energy (MET). Using this study, we derive constraints on the masses of the VLQ and dark matter from existing data. Although similar constraints can be obtained from other *squark* or VLQ searches, they are less stringent in comparison.

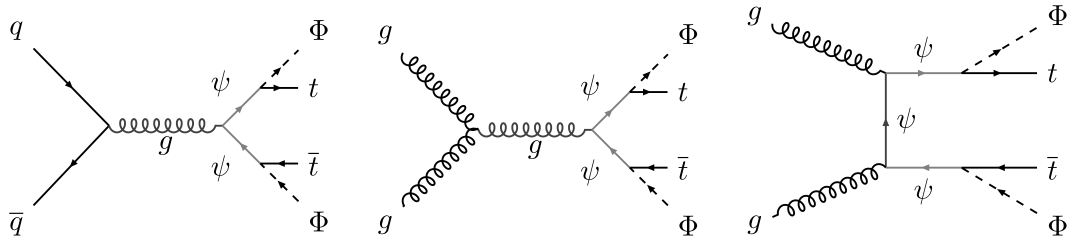


Figure 5: Feynman diagrams corresponding to VLQ production ($t\bar{t} + \not{E}_T$ signal) at the LHC.

The dominant channels of VLQ production (and subsequent decay to top quark and DM) at the LHC are shown in Fig. 5. The model implementation is done using `FeynRules`. The MC events are generated in `MG5_aMC` [97] and the events are showered using `Pythia8` [98]. The showered events are fed into `CheckMATE2` [99] (build upon `Delphes3` [100] and `Fastjet3` [101–103]). `CheckMATE2` uses CL$_s$ method [104] to determine the 95% C.L. exclusion limits. The events are generated at different $\left(m_\psi, m_\Phi\right)$ benchmarks. The 95% C.L. exclusion limit is shown in Fig. 6. The different signal regions considered for the analysis are detailed in [96].

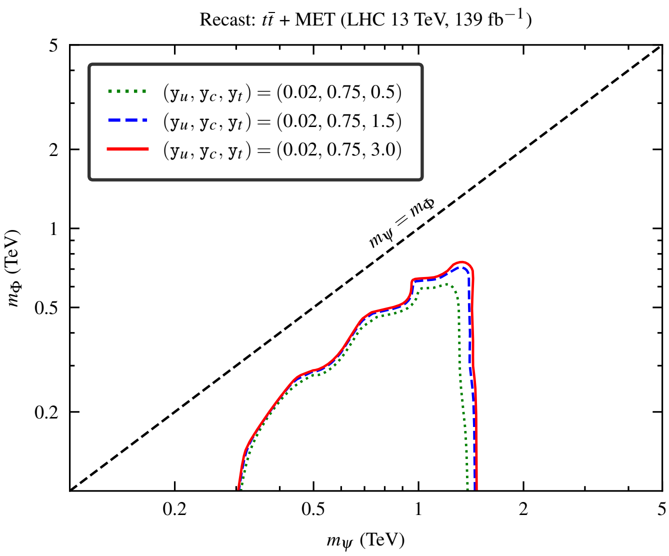


Figure 6: 95% C.L. exclusion lines on the $m_\psi - m_\Phi$ plane from recast of ATLAS top pair + MET analysis [96].

4 Phenomenological analysis

4.1 Flavor observables and model constraints

In the previous section, we discussed how a complex scalar DM with a fermionic mediator impacts the flavor observables of up-type quarks, such as $D^0-\bar{D}^0$ mixing, rare decays of D^0 meson, and top-FCNC decays. In this section, we will show their effect separately and combined on our model parameter space.

$D^0-\bar{D}^0$ mixing:

The short distance effect in D^0 meson mixing is very suppressed in SM, so we have assumed the whole contribution is due to beyond SM effects, as discussed. The observable mass difference will have a dependency on the couplings as: $\Delta m \propto (\mathrm{y_u\,y_c})^2$. Fig. 7a shows the allowed parameter space from $D^0-\bar{D}^0$ mixing in $m_\psi - m_\Phi$ plane. The colored region represents the allowed parameter space. The different color regions correspond to different coupling combinations of $\mathrm{y_u}$ and $\mathrm{y_c}$. The bands represent the allowed parameter space, considering the 1σ error of the observable according to Eq. (3.1). The red, green, orange, and blue color regions correspond to $\mathrm{y_c y_u} = 0.03, 0.02, 0.01$ and 0.005, respectively. The lower mass regions of both DM and mediator are satisfied for smaller coupling values. With increasing the coupling values, we need higher m_Ψ and m_Φ to satisfy the experimental value.

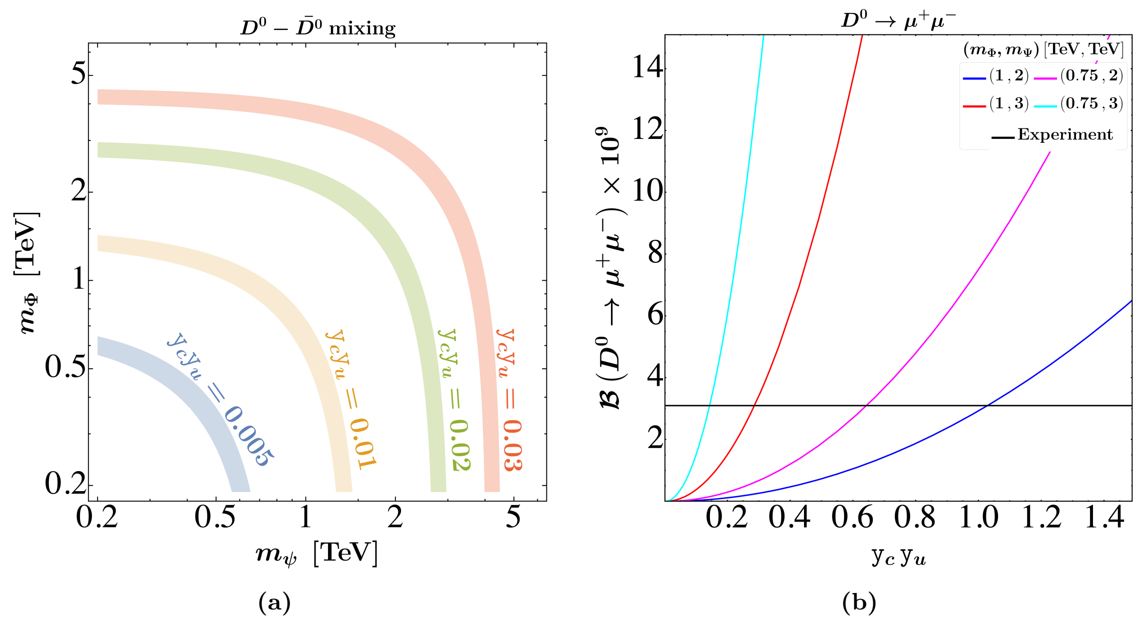


Figure 7: Left: Allowed parameter space from $D^0-\bar{D}^0$ mixing for different combinations of the couplings. The bands are allowed region for 1σ error band. Right: Variation of the branching ratio $\mathcal{B}(D^0 \to \mu^+\mu^-)$ with the couplings $\mathrm{y_c y_u}$, for different combinations of the DM (Φ) and mediator (Ψ) mass. The black line represents the experimental upper limit.

$D^0 \to \mu^+\mu^-$:

The effect of our model on the rare decay $D^0 \to \ell\ell$ is discussed in the previous Section 3.1.2. For rare decays of charm meson, both the short-distance and long-distance effects in the branching ratio are several orders smaller than that of the experimental bound provided, so here also, we take the whole contribution as coming from NP. Our model contributes only to the hadronic part of the decay; the lepton vertex does not receive any BSM corrections. The only new physics contributions appear in the vector current operators $C_{10}^{(\prime)}$. In our case, the decay into an electron pair is further suppressed, as seen from Eq. (3.12). The experimental bound on the electron decay mode is also more relaxed than that for the muon mode. For these reasons, we have only imposed the constraint from $D^0 \to \mu^+\mu^-$. The parameter space consistent with this decay automatically satisfies the constraints from $D^0 \to e^+e^-$ as well.

Fig. 7b shows the variation of the branching ratio of the decay $D^0 \to \mu^+\mu^-$ with the coupling combination $\mathrm{y_c\, y_u}$ since the branching ratio have a dependency $\propto |\mathrm{y_c y_u}|^2$. The branching ratio depends on the coupling $\lambda_{\Phi H}$ through the Feynman diagram shown in the right panel of Fig. 2. However, since this is a scalar-mediated diagram, its contribution is negligible compared to the other diagram. So, this decay process cannot be used to place any bound on the coupling $\lambda_{\Phi H}$. Different colors of Fig. 7b represent various combinations of the DM and mediator masses, while the black line represents the experimental upper limit as given in Eq. (3.7). For a fixed coupling, the branching ratio increases with increasing m_ψ and decreases with increasing m_Φ. Notably, the cyan line shows the most stringent constraint on coupling, corresponding to $m_\Phi = 0.75$ TeV and $m_\psi = 2$ TeV, setting an upper limit $|\mathrm{y_c y_u}| \lesssim 0.12$. Unlike the observable D^0 mixing, where there are measurements, here we only have an upper bound on the branching ratio, allowing any region with couplings below this constraint. Between the mixing and leptonic decays of the D^0 meson, the more stringent bound comes from the $D^0 - \bar{D}^0$ mixing process. The region allowed by mixing is already consistent with the constraint from the $D^0 \to \mu^+\mu^-$ decay. Therefore, in the combined analysis, we do not consider the $D^0 \to \mu^+\mu^-$ constraint separately.

Top-FCNC decays:

This model can also affect top quark FCNC decays, i.e., $t \to c(u)X$ with $X = \gamma, g, Z, h$, through one-loop processes. Among these, the most significant contributions from our model appear in the $t \to c(u)Z$ and $t \to c(u)h$ decay channels. Since these processes involve two up-type quarks, the branching ratios for $t \to c(u)X$ scale as:

$$\Gamma_{t\to c(u)Z,h} \propto |\mathrm{y}_t\, \mathrm{y}_{c(u)}|^2.$$

The corresponding expressions for the branching ratios are given in Eq. (3.16). Note that the branching ratios are computed in the limit $m_{q_j} \to 0$. We have verified that including the mass of the final-state quark does not lead to any significant numerical change.

Unlike the neutral meson mixing process, here we only have an upper bound on the branching ratio, which translates into an upper limit on the allowed parameter space rather

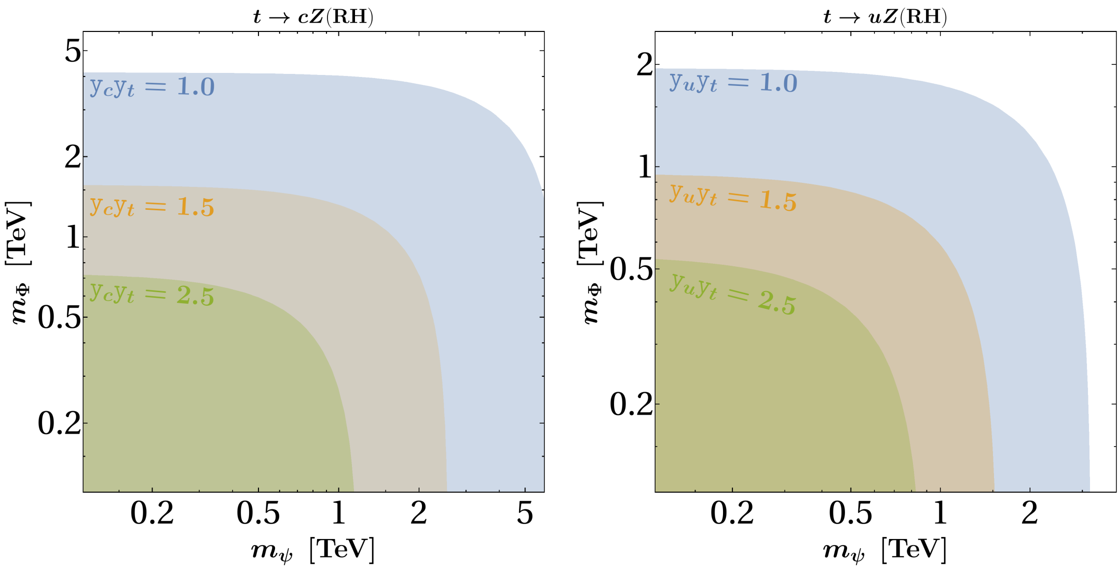


Figure 8: In the left (right) panel, the shaded regions depict the allowed parameter space constrained by the decays $t \to c\ (u)Z$, corresponding to different combinations of coupling constants.

than a band. For the decay involving the Z boson, the experimentally provided upper limits are given separately for the cases where the effective couplings are purely left-handed (LH) or right-handed (RH). The experimental branching ratios for the decays $t\bar{c}(u)Z$ and $t\bar{c}(u)h$, as reported by ATLAS [79] and CMS [76] and summarized in Table 2, indicate that the branching ratio for the decay involving the up quark less than that involving the charm quark. We have performed the analysis for both cases and, in what follows, we present the results for the scenario that yields the more stringent constraint. As mentioned earlier, the dependence of the branching ratio on the mass of the final-state light quark is negligible and does not affect the parameter space. Therefore, the branching ratios for these two channels differ only by the coupling of the DM-mediator to the u and c quarks.

Fig . 8 shows the allowed parameter space in the mediator–DM mass plane (m_Ψ–m_Φ). The left panel corresponds to the decay $t \to cZ$, while the right panel shows results for $t \to uZ$. The parameter space shown is allowed by the experimental upper bounds on the branching ratios, assuming purely right-handed effective couplings, as indicated by the label "RH" in both plots. Since the branching ratio depends on the product of couplings $\mathrm{y}_{c(u)}\,\mathrm{y}_\mathrm{t}$, and not on the individual values, the plots are shown for different values of the parameters $\mathrm{y}_\mathrm{t}\,\mathrm{y}_\mathrm{c}$ and $\mathrm{y}_\mathrm{t}\,\mathrm{y}_\mathrm{u}$, corresponding to the decays $t \to cZ$ and $t \to uZ$, respectively. As only upper bounds are available, the allowed parameter space becomes more relaxed for smaller coupling values. We have illustrated this for three different coupling combinations: $\mathrm{y}_\mathrm{t}\,\mathrm{y}_{c(u)} = 1.0, 1.5$, and 3.0. For the $t \to uZ$ decay, the region $m_\Phi \leq 2$ TeV and $m_\Psi \leq 3$ TeV is allowed for $\mathrm{y}_\mathrm{t}\mathrm{y}_\mathrm{u} = 1.0$, which shrinks with increasing coupling values. For example, when $\mathrm{y}_\mathrm{t}\,\mathrm{y}_\mathrm{u} = 2.5$, the allowed region reduces to $m_\Phi \leq 0.5$ TeV and $m_\Psi \lesssim 0.8$ TeV. The bounds are slightly more relaxed for the $t \to cZ$ decay for the same coupling combinations. For instance, with $\mathrm{y}_\mathrm{t}\,\mathrm{y}_\mathrm{c} = 2.5$, we obtain $m_\Phi \leq 0.7$ TeV and $m_\Psi \leq 1.2$ TeV.

For both decays, we have shown the allowed region only for the "RH" case. In our scenario, the left-handed contribution to the loop is highly suppressed. We have checked that even with a large coupling, $\mathrm{y_c\,y_t} = 3.5$, the entire parameter space remains allowed. So, for the same coupling, the region allowed by "RH" is already covered by "LH" as well. Therefore, we do not show the corresponding plots explicitly.

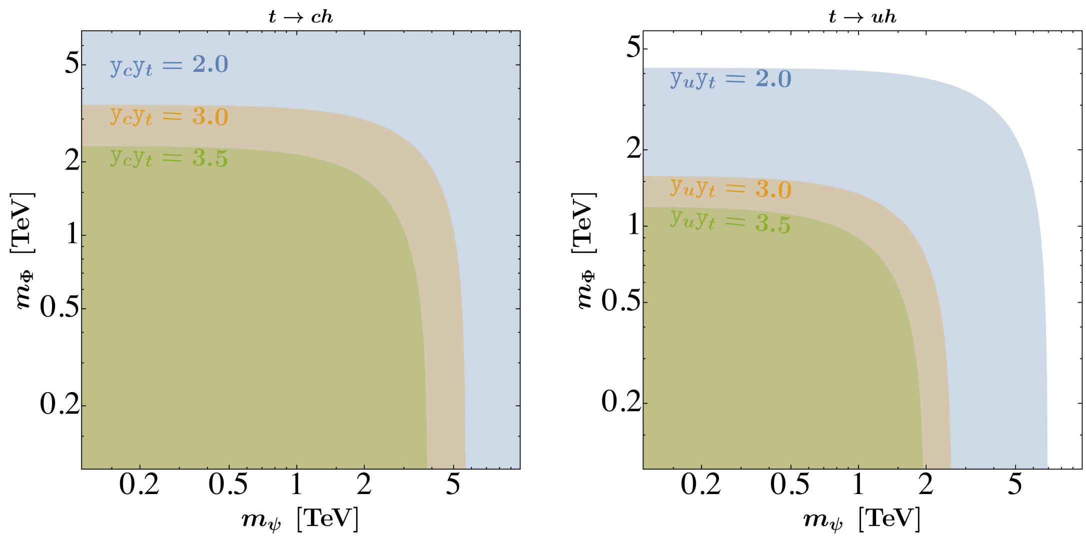


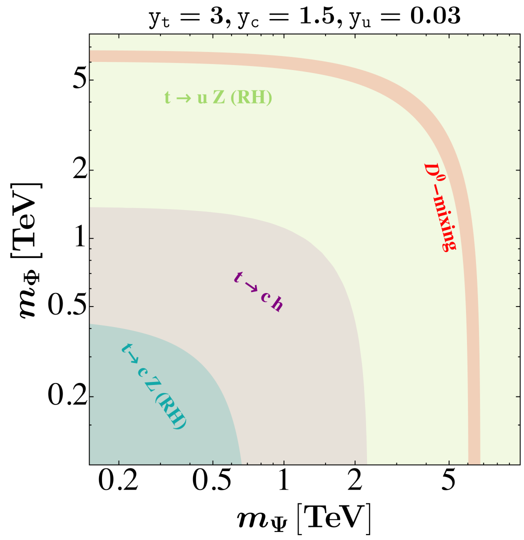
Figure 9: In the left (right) panel, the shaded regions represent the allowable parameter space constrained by the decays $t \to c\ (u)h$, corresponding to various combinations of coupling constants, while fixing $\lambda_{\Phi\mathrm{H}} = 10^{-3}$.

Fig . 9 shows the allowed parameter space from the decays $t \to c(u)h$. The experimental bounds on these decays are given in Table 2. The different colored regions represent the allowed parameter space for various coupling combinations. The branching ratio for $t \to c(u)h$ depends on the coupling as $\Gamma_{t\to c(u)h} \propto \mathrm{y}^2_{c(u)}\mathrm{y}^2_t$, same as $t \to c(u)Z$ and we present the results as a function of the product coupling $\mathrm{y}_{c(u)}\mathrm{y}_t$. As seen in Table 2, the experimental upper limit on the branching ratio is more stringent for the $t \to uh$ process than for $t \to ch$. Consequently, for the same values of couplings to the charm and up quarks, the $t \to uh$ process imposes stronger constraints on the parameter space. This distinction is clearly visible in Fig . 9, where the left panel corresponds to constraints from $t \to ch$ and the right panel from $t \to uh$. Here we have plotted for few sets of the product couplings $\mathrm{y_t y}_{c(u)} = (2.0, 3.0, 3.5)$. For $\mathrm{y_c y_t} = 2.0$, the entire parameter space remains allowed by the $t \to ch$ decay. In contrast, the same coupling combination $\mathrm{y_u y_t} = 2.0$ excludes a portion of the high-mass region of the scalar $\Phi \geq 4$ TeV and $\Psi \geq 7$ TeV. Thus, while this coupling does not constrain the parameter space for $t \to ch$, it does for $t \to uh$. A similar pattern appears at $\mathrm{y_c y_t} = 3.5$, where only the region $m_\psi \geq 4$ TeV is excluded, while for $\mathrm{y_u y_t} = 3.5$, a larger portion of the parameter space is ruled out, i.e., $m_\Phi \geq 1.3$ TeV and $m_\psi \geq 1.9$ TeV.

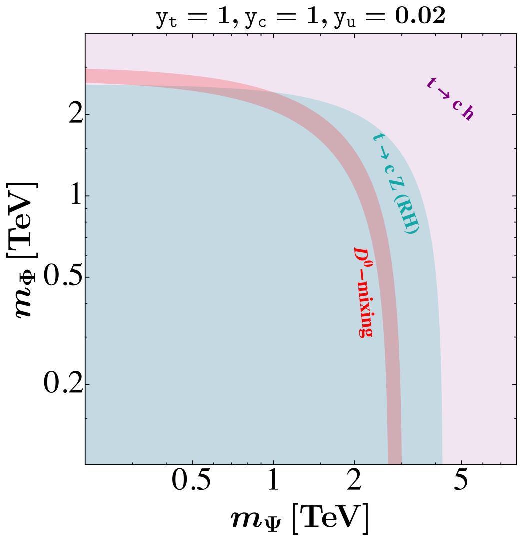
To summarise the parameter space constraints from top-FCNC decays, we find that, for the same values of the couplings, the decays $t \to c(u)Z$ impose stronger constraints than $t \to c(u)h$. Moreover, for equal values of $\mathrm{y_c}$ and $\mathrm{y_u}$, the decay $t \to uX$ (with $X = Z, h$) is more constraining than $t \to cX$. From the previous section, we saw that satisfying the D^0–$\bar{D}^0$ mixing bound requires the product $\mathrm{y_c y_u}$ to be very small, of the order $\lesssim \mathcal{O}(10^{-2})$.

Therefore, to simultaneously satisfy the bounds from top-FCNC decays and mixing, we require $\mathrm{y_c y_u} \sim \mathcal{O}(10^{-2})$ and $\mathrm{y_t y_c(y_u)} \lesssim \mathcal{O}(1)$. This can be achieved either by taking all three couplings to be small, $\lesssim \mathcal{O}(10^{-1})$, or by taking $\mathrm{y_u} \sim \mathcal{O}(10^{-2})$ and $\mathrm{y_c, y_t} \lesssim \mathcal{O}(1)$. After discussing other constraints from dark sectors and colliders, we will use a few suitable benchmarks to do our analysis.

Unified parameter space from all Flavor-Changing processes:



(a)



(b)

Figure 10: The shaded region represents the allowed parameter space constrained by observables from $D^0-\bar{D}^0$ mixing and top-FCNC decays.

In the previous paragraphs, we discussed the bounds from various FCNC processes separately. We found that $D^0-\bar{D}^0$ mixing imposes a very stringent constraint on the parameter space, whereas the bounds from top-FCNC decays are comparatively weaker, resulting in a more relaxed parameter space. Here we will try to find out the common parameter space that is allowed by all the available present bounds. Fig. 10 shows the allowed regions in the DM–mediator mass plane (i.e., the m_Φ–m_ψ plane), where the colored areas correspond to the parameter space allowed by the respective decay processes, indicated by labels of the same color. We show the constraints from the decays $t \to ch$, $t \to cZ$, $t \to uZ$, and $D^0-\bar{D}^0$ mixing. Other processes, such as $D^0 \to \ell^+\ell^-$, $t \to c(u)Z$ (LH), and $t \to uh$, are not shown, as they provide much weaker constraints, as discussed earlier. The plots are presented for two different benchmark scenarios based on different choices of couplings.

Fig. 10a shows the parameter space for the set of the Yukawa couplings: $\mathrm{y_u} = 0.03\,, \mathrm{y_c} = 1.5$ and $\mathrm{y_t} = 3$. The yellow region is for $t \to uZ(RH)$ whereas the sea-green, purple denote red colors are for the allowed region for $t \to cZ(RH), t \to ch$ and $D^0-\bar{D}^0$ mixing process, respectively. The region $m_\Phi \leq 0.40$ TeV and $m_\psi \leq 0.65$ TeV is allowed from the top-FCNC decays but disallowed from the D^0–$\bar{D}^0$ mixing process. We can reduce the coupling $\mathtt{y_u}$ such a way that all of the processes will be satisfied simultaneously. But as we will see in the later part of this work, the smaller mass region will be disallowed from the collider bound. So this will not be a suitable benchmark for our analysis.

Fig. 10b corresponds to the coupling values $\mathtt{y_u} = 0.02$, $\mathtt{y_c} = \mathtt{y_t} = 1$. Since $\mathtt{y_c}\,\mathtt{y_t} = 1$ in this case, a large portion of the parameter space is allowed, extending up to $m_\Phi \leq 2.5$ TeV and $m_\Psi \leq 4$ TeV. The red band indicates the region allowed within the 1σ experimental bound from the $D^0 - \bar{D}^0$ mixing observable. The region consistent with all processes lies within $m_\Phi \leq 2.5$ TeV and 1 TeV $\leq m_\Psi \lesssim 3$ TeV, with a strong mutual correlation between the two masses.

4.2 Dark Matter analysis

The relic density of DM is calculated numerically by solving the Boltzmann Equation (BEQ) corresponding to the CSDM using `micrOMEGAs` [105]. The annihilation and co-annihilation channels that dominantly contribute to the DM number density evaluation are shown in Fig. 18. The relic density of DM is inversely related to the annihilation cross-section. It depends on the initial and final state particle masses, which determine whether the process will be contributing or not, and the annihilation rate is directly proportional to the couplings. To analyze the dependence of the model parameters (see Eq. (2.2)) on the dark matter relic abundance and its direct detection prospects, it is necessary to constrain certain parameters by making specific choices. For the analysis presented in Fig. 11, all parameters have been fixed except for m_Φ and m_ψ (written in terms of $\delta m = m_\psi - m_\Phi$), which are varied within the range of approximately 100 GeV to 10 TeV, subject to the condition $m_\psi > m_\Phi$. The different colored lines illustrate the variation of $\mathtt{y_t}$. The grey-shaded region in the lower plane represents the parameter space excluded by the latest constraints from the LUX-ZEPLIN experiment on the spin-independent dark matter-nucleon scattering cross-section.

Fig. 11 represents the parameter space allowed by relic abundance constraints in the $m_\Phi - \delta m$ (top) and $m_\Phi - \sigma^{\rm eff}_{\Phi \rm N}$ (bottom) planes. The left and right panels correspond to different coupling choices: $\mu_3 = m_\Phi/3$ and $\mu_3 = 2m_\Phi$, respectively. To systematically investigate the contributions of annihilation and co-annihilation processes to the dark matter relic abundance, the dark matter mass range is categorized into three regimes based on its mass splitting with the top quark ($m_{\rm t}$). The effects of two couplings ($\mathtt{y_u}$ and $\lambda_{\Phi\rm H}$) are minimized, while $\mathtt{y_t}$ is varied across three distinct values along with a fixed $\mathtt{y_c}$, ensuring that dark matter remains consistent with direct detection constraints.

- `First` ($m_\Phi < m_{\rm t}$): If the DM particle mass is significantly smaller than $m_{\rm t}$, the relic density dynamics in this regime are predominantly governed by the Higgs portal s-channel and ψ-mediated t-channel interactions, $\Phi\Phi^* \to c\bar{c}$. However, in the limit of a very small $\mathtt{y_c}$, and effectively in the absence of ψ, the model reduces to a minimal CSDM scenario. Here, the choice of $\lambda_{\Phi\rm H}$ is very small, rendering its effect subdominant in Fig. 18k. However, if $\mathtt{y_c}$ is sufficiently larger than the other couplings, then $\Phi\Phi^* \to c\bar{c}$ (Fig. 18m) becomes the dominant process when $m_\Phi > m_\psi/2$, while $\Phi\Phi \to c\bar{\psi}$ (Fig. 18t) dominates when $m_\Phi < m_\psi/2$. As δm decreases, *i.e.*,

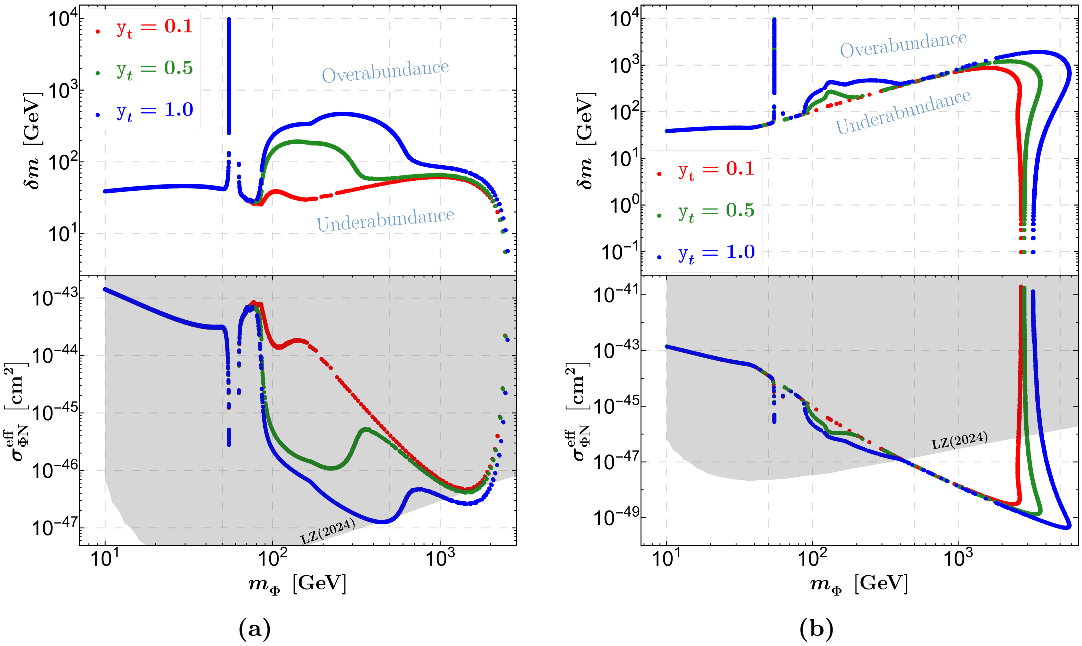


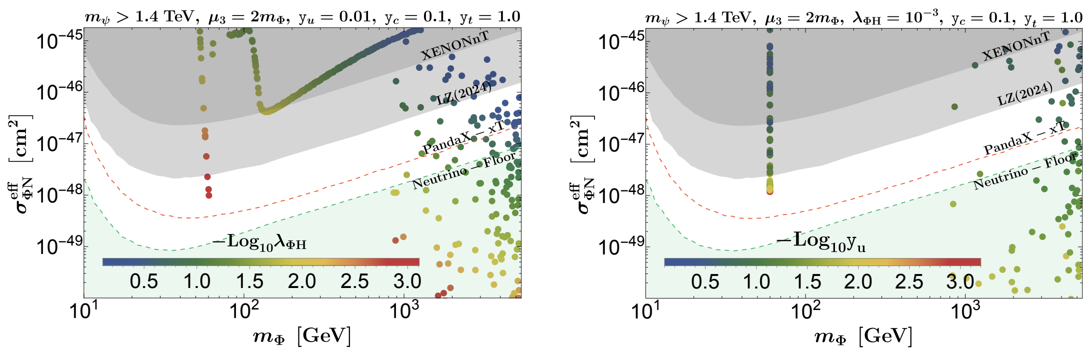
Figure 11: The left (right) figure represents the relic allowed parameter space where $\delta m = m_\psi - m_\Phi$, $\mathrm{y_u} = 10^{-2}$, $\mathrm{y_c} = 0.5$, $\lambda_{\Phi\mathrm{H}} = 10^{-2}$, $\mu_3 = m_\Phi/3$ $(2m_\Phi)$.

as m_ψ becomes smaller, the relic density gradually decreases. Consequently, for 10 GeV $< m_\Phi <$ 50 GeV, the process $\Phi\Phi^* \to c\bar{c}$ predominantly contributes to the observed DM relic density. For these two benchmark scenarios, around the Higgs resonance (with $m_\Phi >$ 50 GeV), the only viable channels to achieve the correct relic density are $\Phi\Phi^* \to b\bar{b}$ below $m_\mathrm{h}/2$ and $\Phi\Phi^* \to c\bar{\psi}$ above $m_\mathrm{h}/2$. A similar reasoning applies to a more underabundant regime for smaller δm. However, this regime is excluded by the spin-independent direct detection limit on DM-nucleon scattering from the LZ-2024 experiment (see the gray-shaded region in the bottom panel of each figure) for this choice of parameters.

- `Second` ($m_\Phi \sim m_\mathrm{t}$): In this regime, the process $\Phi\Phi \to c\bar{\psi}$ dominantly contributes to the relic abundance. As $\mathrm{y_t}$ increases, the process $\Phi\Phi^* \to c\bar{t}$ (Fig. 18m) begins to contribute, leading to a decrease in the relic abundance, which is compensated by increasing m_ψ, *i.e.,* δm. However, when m_Φ exceeds m_t, the contribution from $\Phi\Phi^* \to c\bar{t}$ becomes subdominant, and subsequently, decreasing δm yields the correct relic abundance.

- `Third` ($m_\Phi > m_\mathrm{t}$): If m_Φ is much larger than m_t, the processes $\Phi\Phi \to c\bar{\psi}$ (dominant) and $\Phi\Phi \to t\bar{\psi}$ (sub-dominant) contribute to the relic density. Consequently, with the enhancement of $\mathrm{y_t}$, a slight increase in δm is required to adjust the relic abundance. However, if the DM mass is sufficiently large ($m_\Phi >$ 1 TeV), co-annihilation channels such as $\Phi\bar{\psi} \to$ SM SM and $\psi\bar{\psi} \to$ SM SM begin to dominate over DM self-annihilation. In this regime, δm plays a crucial role in determining the relic density, as reducing the mass splitting between co-annihilating partners enhances the annihilation cross-section. At very high masses ($m_\Phi > 6$ TeV), both self-annihilation and co-annihilation channels become inefficient in achieving the correct relic density, resulting in an overabundant DM scenario. Furthermore, the enhancement of μ_3 leads to an increase in the annihilation cross-section, which can be compensated by an increase in either m_ψ or δm, as clearly illustrated in both plots of Fig. 11.

In Fig. 12, we show the relic-allowed parameter space in the $m_\Phi - \sigma^{\rm eff}_{\Phi \rm N}$ plane. The rainbow color bar represents the variation of $\lambda_{\Phi \rm H}$ (left) and $\rm y_u$ (right), while other parameter values are given at the top of the figure. The gray-shaded regions are excluded by the respective observations, and the green-shaded regions indicate the neutrino floor, where distinguishing between DM and neutrinos is very difficult in the DD experiment. The relevant Feynman diagrams for SI DM-nucleon scattering are shown in Fig. 4. The



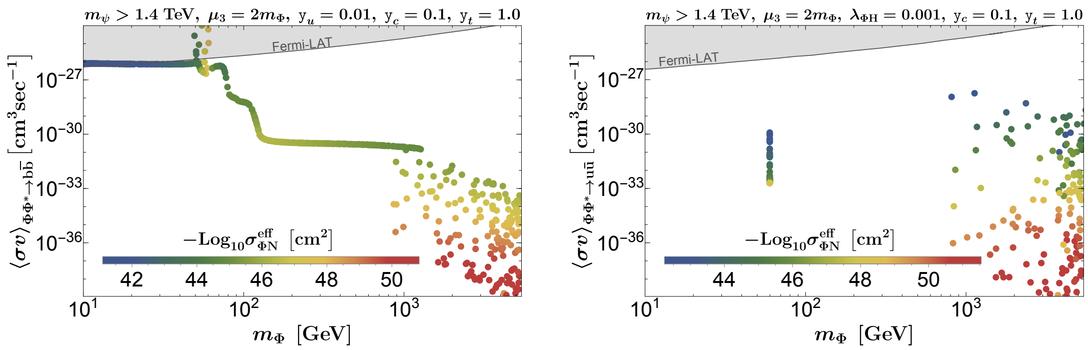
(a)

(b)

Figure 12: Figures represent the relic allowed parameter space in the $m_\Phi - \sigma^{\rm eff}_{\Phi \rm N}$ plane while the rainbow colorbar show the variation of $\lambda_{\Phi \rm H}$ (left) and $\rm y_u$ (right). The gray-shaded regions are excluded by the respective DD experiments mentioned in the figure inset.

cross-section $\sigma^{\rm eff}_{\Phi \rm N}$ has an implicit dependency on the parameters $\{m_\Phi,\ m_\psi,\ \lambda_{\Phi \rm H},\ \rm y_u\}$. Previously, we discussed the dependency of this parameter in the relic context while keeping $\lambda_{\Phi \rm H}$ and $\rm y_u$ fixed. Here, however, these couplings are varied and represented by the color bar. Since these couplings are directly proportional to $\sigma^{\rm eff}_{\Phi \rm N}$, smaller values ($\lambda_{\Phi \rm H} < 1$ and $\rm y_u < 0.1$) are allowed by LZ-2024 while still adhering to the relic constraint. In this regime ($m_\Phi > 1$ TeV), co-annihilation channels become more effective, leading to a relaxation of these couplings. As a result, we obtain a larger relic density and an expanded DD-allowed parameter space in the higher DM mass regime.

The non-observation of DM could face strong constraints from indirect searches like Fermi-LAT, H.E.S.S., and Planck, among others. Here, we are using only the Fermi-LAT data, which puts stringent limits on DM annihilation into the bottom and up quark pairs, as the rate for these processes is not substantially small enough to adjust the DM relic. In Fig. 13, we represent the relic-allowed parameter space. The DM annihilation cross-section gradually decreases with m_Φ beyond the Higgs resonance. However, around the Higgs resonance, some points are excluded by the Fermi-LAT limit on the DM annihilation cross-section $m_\Phi - \langle \sigma v \rangle_{\Phi\Phi^* \to b\bar{b}}$. In Fig. 13b, we vary $\rm y_u \geq 0.1$ so that the entire parameter



(a) **(b)**

Figure 13: The Fermi-LAT observed bounds are illustrated on the relic allowed parameter-space in $m_\Phi - \langle\sigma v\rangle_{\Phi\Phi^*\to b\bar{b}}$ (*left*) and $m_\Phi - \langle\sigma v\rangle_{\Phi\Phi^*\to u\bar{u}}$ (*right*) plane, by the thick gray line.

space is relaxed from the Fermi-LAT limit. Here, the DM annihilation cross-sections $m_\Phi - \langle\sigma v\rangle_{\Phi\Phi^*\to b\bar{b}}$ and $m_\Phi - \langle\sigma v\rangle_{\Phi\Phi^*\to u\bar{u}}$ are proportionally related to $\lambda_{\Phi\mathrm{H}}$ and $\mathrm{y_u}$, similar to the DD cross-section. The direct ($\sigma^{\rm eff}_{\Phi\mathrm{N}}$) and indirect cross-sections decrease with decreasing couplings, as indicated by the rainbow color bar in terms of $\sigma^{\rm eff}_{\Phi\mathrm{N}}$. In Fig. 13a (13b), the variation of $\sigma^{\rm eff}_{\Phi\mathrm{N}}$ is illustrated as a function of $\lambda_{\Phi\mathrm{H}}$ ($\mathrm{y_u}$), while other parameters remain fixed, as specified in the figure inset. Therefore, a better coupling choice should be used for analyzing the explained flavor data and predicting the collider signal, considering very small $\lambda_{\Phi\mathrm{H}}$ and $\mathrm{y_u}$ couplings. This ensures that the parameter space remains relic-density allowed while also staying within the direct detection (DD) and indirect detection (ID) bounds.

4.3 VLQ search at future muon colliders

Traditionally, hadron colliders have been the primary facility for searching for VLQs (discussed in Section 3.3), given its connection to the quark sector and large dataset from proton-proton collisions. However, searching for VLQs connected to the dark sector at hadron colliders faces significant challenges, particularly in reconstructing final states involving MET in presence of complex QCD backgrounds. The overwhelming hadronic environment at the hadron leads to substantial background contamination, making it difficult to isolate new physics signals from SM processes. In contrast, muon colliders present a cleaner environment with distinct advantages for such searches. The reduced QCD background in lepton collisions allows for better signal discrimination with distinct observables. Secondly, muon colliders offer a significantly enhanced ability to probe NP scenarios and exotic decays due to more precise energy control and suppressed initial-state radiation compared to hadron colliders, significantly improving searches involving invisible particles in the final states. Moreover, muon colliders can achieve TeV-scale partonic collision energies with high luminosities, making them well-suited for direct searches of heavy VLQs that may be difficult to detect at the LHC.

In this section, we study VLQ pair production at the muon collider with CM energy, $\sqrt{s} = 10$ TeV with an integrated luminosity of 10 ab^{-1}. We focus on the $t\bar{c}$ + missing energy

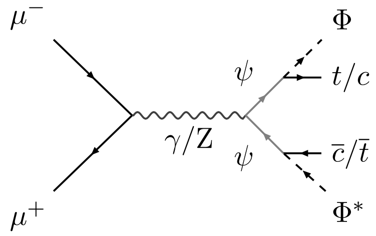


Figure 14: Feynman diagrams correspond to VLQ production ($t\bar{c}/\bar{t}c$+$\not{E}$ signal) at lepton colliders.

final state, where one VLQ decays to a top quark and DM, and the other VLQ deacys to a charm quark and DM. The feynman diagram for the signal process is shown in Fig . 14. At multi-TeV muon colliders, collisions take place at a CM energy surpassing that achieved by the current LHC (parton level) or potential future e^+e^- colliders. Consequently, it is anticipated that jets which are closely clustered together in these collisions, will exhibit collimation, effectively coalescing into a singular, consolidated jet. A notable illustration of this behavior can be observed in the jets stemming from the hadronic decay processes of top quarks, Higgs or W/Z bosons, where they converge to form a single "top", "Higgs" or "W/Z" jet [106, 107]. Consequently, our signal process reduces to a dijet plus missing energy signature, with one 'heavy' jet (resulting from top decay). We further impose the selection criteria of no detected photons or leptons in the final state. The event generation follows the same approach as described in Section 3.3. The detector simulation is done using the default `Delphes` card in `MG5_aMC`, which takes care of detector resolution and tagging efficiencies. The jet clustering is done using the `anti-kt` algorithm [101–103]. The jet radius is chosen to be 0.5 and the minimum p_T of the jet is 20 GeV. For the analysis, we choose the benchmark point,

$$\text{BP}: \quad m_\Phi = 800 \text{ GeV}, \quad m_\psi = 1575 \text{ GeV}, \quad (\mathrm{y_u}, \mathrm{y_c}, \mathrm{y_t}) = (0.02, 0.75, 1.5), \tag{4.1}$$

which satisfy the observed DM relic density and abide by the different constraints discussed in previous sections.

The major SM background processes are tbW, $\nu\bar{\nu}WW$, Wjj, $\nu\bar{\nu}ZZ$, $t\bar{t}Z$ and Zjj. In addition to the NP signal, $t\bar{c}\Phi\Phi^*$, there are NP backgrounds, $t\bar{t}\Phi\Phi^*$ and $c\bar{c}\Phi\Phi^*$, which should also be taken into account. Some kinematic variables for the NP signal process and the SM backgrounds are plotted in Fig . 15. The definition of the variables are as follows:

- M_{j_1} and M_{j_2} are masses of the heavier and lighter jet respectively. Jet mass is defined as the invariant mass of the reconstructed particle constituents of the the jet.
- The missing energy, $\not{E}$ of an event is defined as $\not{E} = \sqrt{s} - \sum_i E_i$, where $\sum_i E_i$ is the sum of the energy of the visible final state particles.
- The invariant mass of the jets, M_{jj} is defined as $M_{jj} = \sqrt{(p_{j_1} + p_{j_2})^2}$, where p_{j_1} and p_{j_2} are the 4-momenta of the heavier and lighter jet respectively.

In order to segregate the NP signal from the background processes, we apply sequential kinematic cuts (on top of selection and detector cuts discussed earlier):

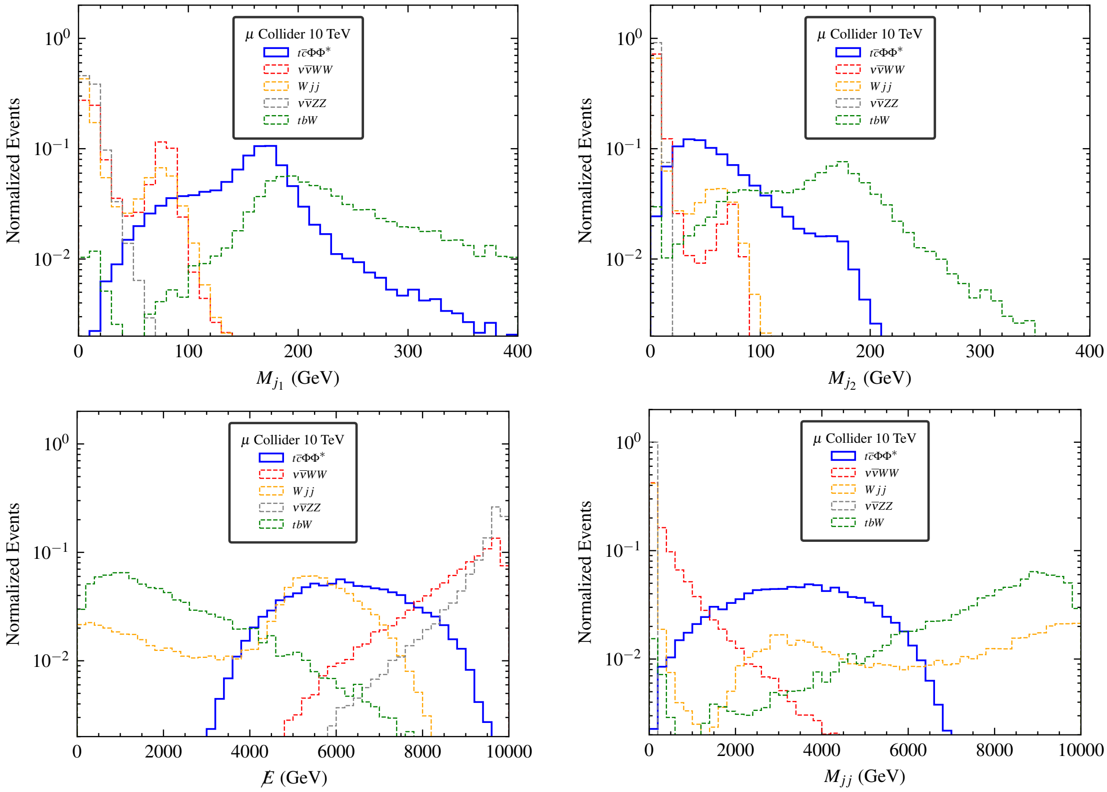


Figure 15: Kinematic distributions: *Top left:* Mass of heavier jet, M_{j_1}, *Top right:* Mass of lighter jet, M_{j_2}, *Bottom left:* Missing energy, $\not{E}$, *Bottom right:* Invariant mass of jet pair, M_{jj}, corresponding to signal and major background processes at the 10 TeV muon collider.

- $\mathcal{C}_1$: 125 GeV $< M_{j_1} <$ 200 GeV,
- $\mathcal{C}_2$: $M_{j_1} <$ 150 GeV,
- $\mathcal{C}_3$: 3000 GeV $< \not{E} <$ 8000 GeV,
- $\mathcal{C}_4$: 2000 GeV $< M_{jj} <$ 7000 GeV.

The M_{j_1} distribution for the signal is expected to peak at top quark mass, $m_t \sim 175$ GeV. We impose a wide cut, $\mathcal{C}_1$ owing to the difficulty in the reconstruction of constituents of a boosted jet. This effectively reduces background without top quarks in the final state. The next cut, $\mathcal{C}_2$ is imposed to backgrounds containing more that a single top quark in the final state like the background, tbW, where the dominant contribution comes from on-shell $t\bar{t}$ process. The cut windows, $\mathcal{C}_3$ and $\mathcal{C}_4$ are chosen to reduce the remaining background while losing minimum signal in this process. These cuts are selected in order the optimize the signal-background discrimination for our chosen benchmark. The detailed cutflow is shown in Table 3 for the signal benchmarks and the backgrounds. The backgrounds $t\bar{t}Z$ and Zjj do not survive the cutflow and hence not mentioned in the table. The signal significance is defined as:

$$\text{Significance} = \frac{\sigma_s \times \mathfrak{L}_{\text{int}}}{\sqrt{\sigma_b}}\,, \tag{4.2}$$

where, σ_s and σ_b are total cross sections of the signal and backgrounds respectively, after applying the sequential cuts. $\mathfrak{L}_{\rm int}$ is the integrated luminosity of the collider, chosen to be 10 ab^{-1} here. We observe that both the signal benchmark has signal significances higher than 5σ, the commonly accepted discovery limit for NP signals [108]. The signal significance for the $t\bar{c}\Phi\Phi^*$ signal is plotted in the $m_\psi - m_\Phi$ plane, for the two different benchmark cases, in Fig. 16.

Processes		$\sigma(\mathcal{C}_1)$ (fb)	$\sigma(\mathcal{C}_2)$ (fb)	$\sigma(\mathcal{C}_3)$ (fb)	$\sigma(\mathcal{C}_4)$ (fb)
SM	$\nu\bar{\nu}WW$	7.955×10^{-1}	7.914×10^{-1}	2.867×10^{-1}	5.249×10^{-2}
	Wjj	5.702×10^{-2}	5.604×10^{-2}	1.640×10^{-2}	1.468×10^{-2}
	tbW	2.205×10^{0}	1.513×10^{0}	7.587×10^{-1}	7.112×10^{-1}
	$\nu\bar{\nu}ZZ$	1.251×10^{-1}	1.251×10^{-1}	5.041×10^{-2}	2.054×10^{-2}
BP	$t\bar{c}\Phi\Phi^*$	8.425×10^{-2}	8.058×10^{-2}	7.423×10^{-2}	6.882×10^{-2}
	$t\bar{t}\Phi\Phi^*$	7.883×10^{-2}	5.771×10^{-2}	5.246×10^{-2}	4.818×10^{-2}
	$c\bar{c}\Phi\Phi^*$	9.310×10^{-3}	9.172×10^{-3}	8.574×10^{-3}	7.933×10^{-3}
	Significance	4.658	5.044	6.853	7.443

Table 3: Cutflow and signal significance for $t\bar{c}\Phi\Phi^*$ signal process along with NP ($t\bar{t}\Phi\Phi^*$ and $c\bar{c}\Phi\Phi^*$) and SM ($\nu\bar{\nu}WW$, Wjj, tbW and $\nu\bar{\nu}ZZ$) backgrounds at the muon collider 10 TeV 10 ab^{-1}. Here, $\sigma(\mathcal{C}_i)$ refers to as the cross section of the corresponding process following sequential cut, $\mathcal{C}_i$.

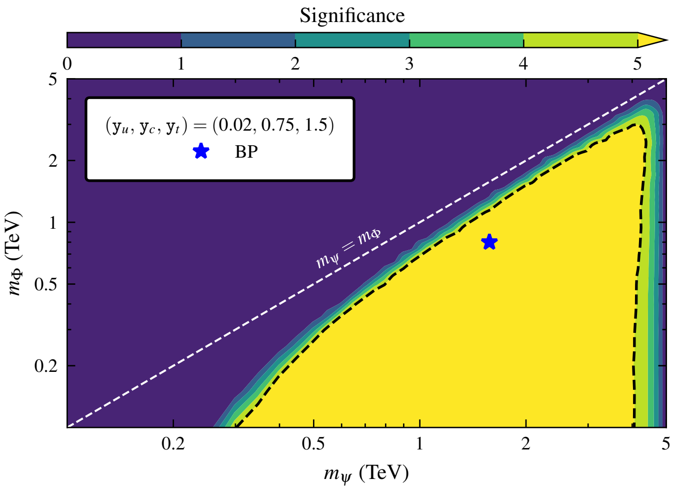


Figure 16: Signal significance for $t\bar{c}\Phi\Phi^*$ signal process at the muon collider 10 TeV 10 ab^{-1} on the $m_\psi - m_\Phi$ plane. The black dashed line corresponds to the 5σ significance limit. The blue star corresponds to benchmark point, BP.

4.4 Combined interpretation

In Fig. 17, we show a combined interpretation of the exclusion limits and future sensitivities in the $m_\psi - m_\Phi$ parameter space for a benchmark value:

$$y_u = 0.02,\ y_c = 0.75,\ y_t = 1.5,\ \mu_3 = 2m_\Phi, \quad \text{and}\ \lambda_{\Phi H} = 10^{-3}.$$

The cyan band represents the region allowed by D^0–$\bar{D}^0$ mixing constraints (see Section 3.1 and 4.1). The light-blue shaded region represents the excluded parameter space from the $t \to cZ$(RH) decay, discussed in Section 4.1. The grey shaded area is excluded by the SI DM-nucleon scattering limits from LUX-ZEPLIN (2024), discussed in Section 4.2. The Magenta shaded region shows exclusion limits from a recast of the $t\bar{t}$+ MET signal at the LHC (13 TeV, 139 fb^{-1}) for this benchmark (see Section 3.3). The blue dashed curve indicates the 5σ discovery reach for the $t\bar{c}+\not{E}$ signal at a future 10 TeV muon collider with an integrated luminosity of 10 ab^{-1} (Section 4.3). The Magenta colored points indicate the parameters are consistent with the observed DM relic abundance, and indirect detection limits from Fermi-LAT and Planck observations (Section 4.2).

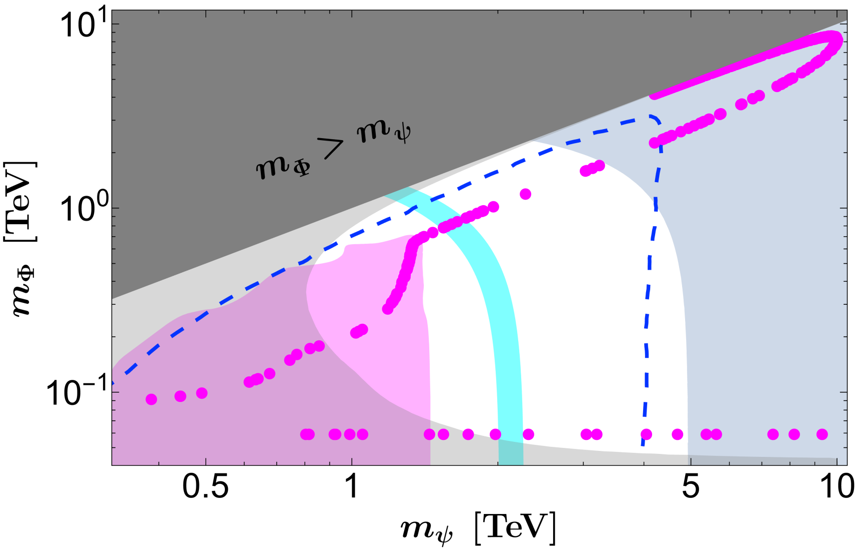


Figure 17: The magenta color points in $m_\psi - m_\Phi$ plane, satisfy the DM relic abundance and also respect the indirect detection limit on DM annihilation. The other model parameters are fixed as: $\mathrm{y}_u = 0.02$, $\mathrm{y}_c = 0.75$, $\mathrm{y}_t = 1.5$, $\mu_3 = 2m_\Phi$, and $\lambda_{\Phi\mathrm{H}} = 10^{-3}$. The cyan-colored band represents the $D^0 - \bar{D}^0$ mixing. The grey, magenta, and light-blue shaded regions are excluded by the LUX-ZEPLIN (2024), LHC, and $t \to cZ$ (RH) decay, respectively. The orange dashed line corresponds to $m_\psi = m_\Phi$. The dark gray shaded region, corresponding to $m_\Phi > m_\psi$, is excluded based on the known properties of DM. The blue dashed line represents the 5σ signal significance at the 10 TeV muon collider with an integrated luminosity of 10 ab^{-1}.

It is important to note that the CSDM is only consistent with the observed relic density, DD, and ID constraints in the regions ($m_\Phi \sim m_\mathrm{h}/2$, $m_\psi \gtrsim 2$ TeV) and ($m_\Phi \gtrsim m_\mathrm{t}$, $m_\psi \gtrsim 1$ TeV). This compatibility arises due to the small values chosen for $\lambda_{\Phi\mathrm{H}}$ and $\mathrm{y_u}$, which are effective for the spin-independent DD and CSDM annihilation cross-sections relevant to indirect detection. In contrast, the relic abundance is mostly influenced by other model parameters such as $\mathrm{y_c}$, $\mathrm{y_t}$, μ_3, m_Φ, and m_ψ, which are less relevant for DM detection apart from collider studies, where $\mathrm{y_t}$ and $\mathrm{y_c}$ are significant along the masses. In the high-mass regime ($m_\Phi \gtrsim 2$ TeV and $m_\psi \gtrsim 4$ TeV), co-annihilation channels become dominant (as discussed in Section 4.2), resulting in nearly degenerate CSDM masses for a fixed m_ψ could achieve the correct relic. This bending kind of behavior arises only in the higher mass regime ($m_\Phi \gtrsim 1$ TeV) due to the following reasons: if the CSDM mass is $\sim m_\psi/2$, then the $\Phi\Phi^* \to q_{\rm u}\bar{\psi}$ channel (Fig. 18t) dominantly contributes to the DM freeze-out and gives a relic-allowed points. However, if we gradually increase m_Φ for a fixed m_ψ, other co-annihilation processes (Fig. 18) start to contribute and yield under-abundant relic points. On the contrary, for $m_\psi/2 \ll m_\Phi \lesssim m_\psi$, the effect of the co-annihilation channels starts to decrease, leading to another correct relic point before entering into the overabundance regime.

Finally, in Fig. 17, we see that the points around $m_\psi \sim 1.5$TeV and $m_\Phi \sim 1$ TeV are consistent with all current experimental limits for this benchmark. However, other benchmark scenarios may also satisfy these constraints and could be probed by future experiments. But the model parameters simultaneously influence dark matter, flavor, and collider phenomenology. Therefore, the viable parameter space is highly constrained and typically requires fine-tuning to simultaneously accommodate all observables.

5 Summary and Conclusion

In this article, we present a simple model that addresses two key issues in particle physics: Dark Matter (DM) and the observables related to FCNC processes. This is accomplished by extending the Standard Model (SM) particle contents with a complex scalar field and a singlet Dirac vector-like quark (VLQ), which couples with the SM up-type quarks due to chosen $\rm U(1)_Y$ hypercharge. In this minimal extension, the complex scalar field acts as the DM candidate, with its stability ensured by an unbroken $\mathcal{Z}_3$ symmetry. The VLQ transforms similarly under this symmetry but is chosen to be heavier than the complex scalar, allowing it to decay into the DM and visible particles.

The study elaborates upon the correct DM relic density allowed parameter space consistent with the current direct detection (DD) and indirect detection (ID) limits. Due to the existence of heavy dark sector VLQ, the depletion is aided by co-annihilation processes whenever the mass splitting remains small. Apart, the $\mathcal{Z}_3$ symmetry allows Φ^3 interaction, which plays crucial role in achieving correct relic density via semi annihilation processes of the type $\Phi\Phi \to \bar{\psi}u$, whenever $2m_\Phi > m_\psi$. Apart, the interaction between the dark sector and visible sector particles is also mediated by the VLQ and Higgs portal interaction, while the latter is suppressed to remain consistent with current DD and ID constraints. The Yukawa couplings between VLQ, DM and SM particles play a crucial role in DM relic as well as FCNC and collider observables.

Existing LHC searches for DM via missing energy imposes strong bounds, pushing both the VLQ and DM particles into the high-mass regime. The most stringent limits come from the observation of $t\bar{t}+$MET at ATLAS, which yields $m_\psi \gtrsim 1.4$ TeV for $(m_\Phi + m_{\rm t}) < m_\psi$.

As the model offers interaction between VLQ, DM and right-handed up-type quarks, it contributes to FCNC processes; we have considered $D^0-\bar{D}^0$ mixing, rare decays of D^0 meson, and top-FCNC decays. The processes related to D^0 depend on the couplings $\rm y_u$ and $\rm y_c$, while top-FCNCs are impacted by all three Yukawa couplings. The most relevant flavor observables, $D^0-\bar{D}^0$ mixing and $t \to cZ$ (RH), significantly impact DM phenomenology and collider search prospects. Similar extensions can be constructed for left-handed quark doublets or right-handed down-type quarks, with appropriate charge assignments for the VLQ and DM. Such variations would introduce additional flavor constraints from the other low-energy observables and open up a direct connection between the DM and the full quark sector. A more comprehensive analysis can be taken up in future in this direction.

We explore the viable parameter space of the model consistent with the constraints outlined in Section 3, and provide predictions in future muon collider experiments operating at $\sqrt{s} = 10$ TeV via $t\bar{c}$ plus missing energy. Missing energy and invariant mass of the jet pair help in signal-background segregation. Due to high VLQ mass, the prospects at the LHC seems difficult, and so is at future electron positron machines. The summary plot (Fig . 17) represents the relic-allowed parameter space for a benchmark scenario consistent with direct, indirect DM search constraints, collider sensitivity, observed $D^0-\bar{D}^0$ mixing data and $t \to cZ$ (RH) exclusion.

In summary, our model presents a minimal yet compelling framework linking DM and flavor physics, where there exists a correlation between both the sectors. It remains consistent with current experimental searches, while the detection seems to be possible at future muon colliders, providing a compelling reason to build such machines to explore BSM physics.

A Feynman diagrams related to DM relic

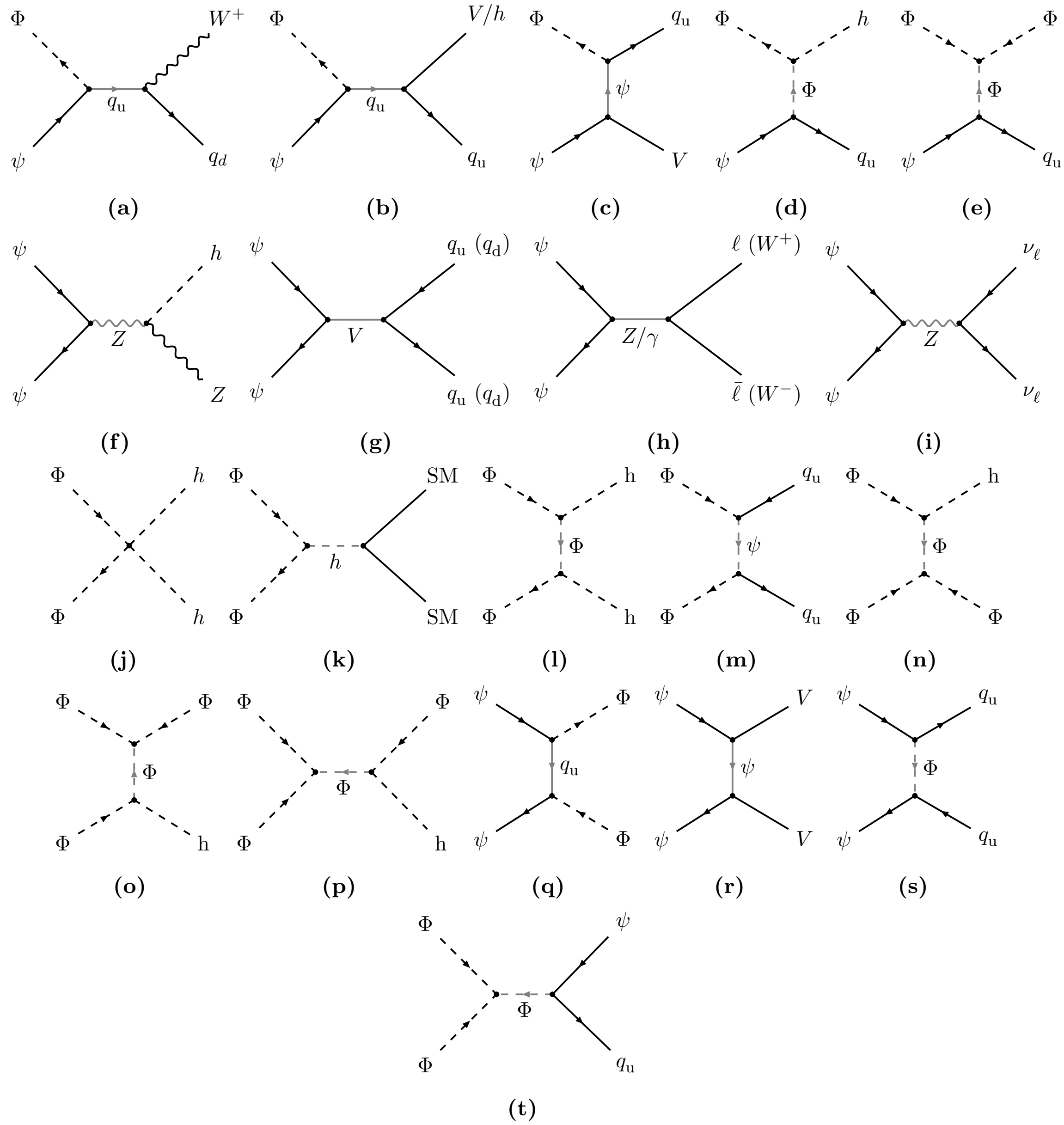


Figure 18: The Feynman diagrams are related to the annihilation and co-annihilation of WIMP ($\psi\ \Phi^* \to q_\mathrm{u}\ V/h$, $\psi\ \Phi \to q_\mathrm{u}\ \Phi$, $\Phi\ \Phi^* \to$ SM SM, $\Phi\ \Phi \to \Phi^*$ h, $\psi\ \bar{\psi} \to \Phi\ \Phi^*$/h Z/q $\bar{\mathrm{q}}$/$\mathrm{q_u}\ \bar{\mathrm{q}}_\mathrm{u}$/$\ell\ \bar{\ell}$/$\nu_\ell\ \bar{\nu}_\ell$/V V/W$^+$ W$^-$) where $V \in \{\gamma,\ Z,\ g\}$, $q_\mathrm{u} \in \{u,\ c,\ t\}$, $q_d \in \{d,\ s,\ b\}$, $\ell \in \{e, \mu, \tau\}$, $\nu_\ell \in \{\nu_e, \nu_\mu, \nu_\tau\}$ and SM $= \{\mathrm{h},\ \mathrm{q_u},\ \mathrm{q_d},\ \ell,\ \mathrm{Z},\ \mathrm{W^\pm}\}$.

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